School of Computer Engineering

Manipal Institute of Technology

Manipal Academy of Higher Education

Manipal, Karnataka, India

Auditing the Faithfulness and Cross-City Consistency of Attention in Multimodal Spatiotemporal Air Quality Index Prediction

Results, Discussion and Inference

ML Lab Project - 20 October 2026

Including same-day estimation and next-day forecasting

Team member	Registration number	Roll number
Saanvi Garg	240905702	68
Sparsh Agrawal	240905708	70
Dhruv Chamria	240905706	69
Neelaksha Sisodiya	240905642	65

Section CSE-C

Faculty guide: Dr Nisha P Shetty

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IV. Results, Discussion and Inference

A. Experimental Results

The completed experiments use distinct chronological test periods. Within each period, comparisons use the same eligible station-days. The 2024 same-day, 2025 next-day, 2026 paired forecast, and 2023 attention results should not be ranked against one another as though they shared one test distribution.

A.1 Metric-wise performance and historical baselines

Table 1 shows that the 2024 same-day Ridge model with AQI history reduced RMSE from 36.197 for persistence to 32.806 AQI points on 28,803 station-days, a 9.4% reduction. Its MAE was 23.001, compared with 23.716 for persistence. Environment-only Ridge had RMSE 63.239. In the separate 2025 next-day test, the three-seed 28-day GRU reduced RMSE from 35.388 for persistence to 32.516 on 30,263 station-days, an 8.1% reduction. Its MAE was 21.857, compared with 22.981 for persistence.

Table 1 Historical performance against persistence. Each task has its own test cohort.

Model	Station-days	MAE	RMSE	R^2
Same-day estimation, 2024				
Ridge with AQI history	28,803	23.001	32.806	0.892
Persistence	28,803	23.716	36.197	0.869
Environment-only Ridge	28,803	45.804	63.239	0.599
Next-day forecast, 2025				
Three-seed GRU ensemble	30,263	21.857	32.516	0.887
Persistence	30,263	22.981	35.388	0.867

MAE and RMSE are in AQI points. Same-day estimation uses target-day environmental observations; next-day forecasting does not.

The paired weekly block-bootstrap RMSE gain was 3.391 AQI points [95% interval: 2.03, 4.67] for 2024 same-day Ridge and 2.872 [2.168, 3.605] for the 2025 GRU. GRU RMSE was lower than persistence in all eight cities in 2025. The nominal 90% station prediction interval covered 95.4% of the observed 2025 targets; this is historical coverage, not a future guarantee.

A.2 Paired 2026 forecast comparison

Weather-assisted XGBoost, the frozen GRU, and persistence were evaluated on exactly the same 20,086 station-days from January-September 2026 (91 of 107 configured stations). Table 2 and Fig. 1 show the metric-wise comparison. XGBoost had the lowest MAE and RMSE, and its RMSE was lower than GRU in each of the eight represented cities. Its paired RMSE gain was 1.437 points over GRU [95% weekly-bootstrap interval: 0.376, 2.614] and 4.398 points over persistence [3.190, 5.715].

Table 2 Paired January-September 2026 forecast comparison.

Model	Station-days	MAE	RMSE	R^2
Weather-assisted XGBoost	20,086	19.345	30.270	0.848
Frozen GRU ensemble	20,086	20.394	31.707	0.833
Persistence	20,086	21.763	34.668	0.800

All three scores use identical rows. MAE and RMSE are in AQI points.

A. Experimental Results (continued)

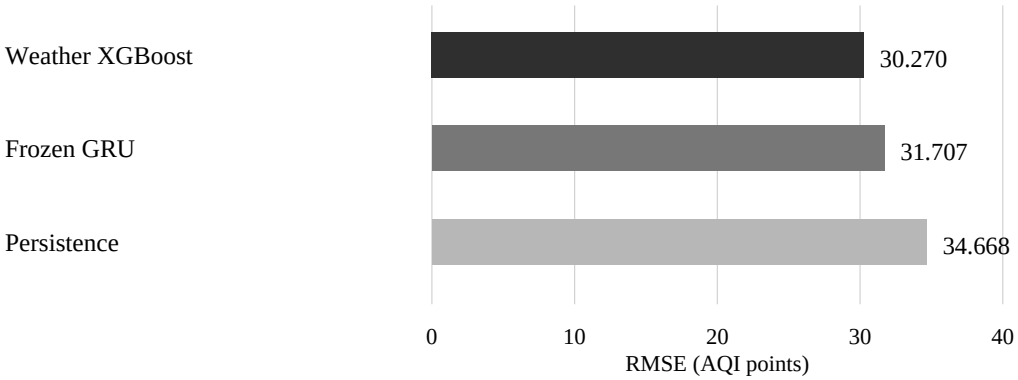


Fig. 1 RMSE on the same 20,086 January-September 2026 station-days. Lower is better; these bars do not compare different test years.

A.3 Abrupt-change and severe-AQI outcomes

Average error did not capture the full operational tradeoff. Of 2,315 observed station-days with an absolute daily AQI change of at least 50, XGBoost detected 422 changes in the correct direction, versus 154 for GRU. It flagged 799 changes overall, versus 268 for GRU (Table 3). XGBoost therefore had higher recall but lower precision.

Table 3 Sharp-change detection on the paired 2026 cohort; threshold = 50 AQI points.

Measure	Weather XGBoost	Frozen GRU
Observed sharp changes	2,315	2,315
Forecast-flagged changes	799	268
Correct directional detections	422	154
Recall	18.23%	6.65%
Precision	52.82%	57.46%

A correct detection must exceed the threshold in the observed direction. Precision is computed over forecast-flagged changes.

For observed AQI of 301-400, the weather model's mean prediction error was -38.51 points; for 401-500 it was -44.36 points. Only 94 observations were in the latter band, so that estimate is less stable. These severe-day errors matter even though the model led the average-error metrics.

A.4 Attention accuracy and faithfulness

The corrected attention study used a separate November-December 2023 chronological test of 5,434 station-days. Within the seven-day sequence family, temporal attention (TA), feature attention (FA), and their combination all had higher RMSE than uniform weighting. Within the spatial-patch family, spatial attention (S-SA) reduced RMSE from 75.328 to 65.836 (Table 4).

A. Experimental Results (continued)

Table 4 Corrected attention experiment, November-December 2023.

Model family	Variant	Station-days	RMSE
Seven-day sequence	Uniform weights	5,434	78.361
Seven-day sequence	Temporal attention (TA)	5,434	84.394
Seven-day sequence	Feature attention (FA)	5,434	82.284
Seven-day sequence	TA + FA	5,434	94.525
Spatial patch	Uniform weights	5,434	75.328
Spatial patch	Temporal attention (S-TA)	5,434	83.641
Spatial patch	Spatial attention (S-SA)	5,434	65.836
Spatial patch	S-TA + S-SA	5,434	75.966

RMSE is in AQI points. Compare variants only within the same model family.

Faithfulness was audited on 300 sampled rows for each corrected FA and S-SA checkpoint. Table 5 reports row-weighted mean Spearman agreement between each ranking and fixed-model input-replacement sensitivity. SHAP exceeded attention under both replacement methods for both models; the size of the gap depended on the replacement method.

Table 5 Ranking agreement with input-replacement sensitivity.

Model	Replacement method	Attention r	SHAP r	Rows
FA	Mean replacement	0.420	0.664	300
FA	Same-city seasonal donor	0.392	0.546	300
S-SA	Mean replacement	0.242	0.419	300
S-SA	Same-city seasonal donor	0.483	0.586	300

Values are row-weighted mean Spearman correlations, not changes in forecast accuracy.

B. Discussion

B.1 Why the average-error models differed

Persistence was a strong baseline in every forecast comparison, showing that recent AQI carries substantial predictive information. The much poorer 2024 environment-only Ridge score, alongside the better history-enhanced Ridge score, supports the importance of an AQI-history anchor in same-day estimation. The 28-day GRU improved on repeating yesterday's value in the 2025 cohort, consistent with useful information in a longer AQI sequence and known calendar/location inputs. That comparison does not isolate which individual GRU input caused the gain.

The 2026 weather-assisted XGBoost model predicts a correction to the previous-day AQI and uses archived forecasts available before the target day. Its lower paired error is consistent with a useful combination of that anchor, nonlinear residual modeling, and forecast-weather context. Because the comparator is a different architecture, this test alone cannot measure a weather-only causal contribution. The observed gain is a model-level result, not proof that any single weather variable explains it.

B.2 Where the gain weakened

The weather candidate flagged roughly three times as many sharp changes as GRU (799 versus 268), which helps explain its higher recall and lower precision. Its underprediction in high-AQI bands shows why a lower overall RMSE can coexist with weak performance on severe episodes. Rare extremes, incomplete station coverage, and different city weights make the pooled result insufficient as an operational safety criterion. Delhi contributed 12,387 of 30,263 rows in the 2025 GRU test (40.9%), so that pooled metric gives Delhi substantial influence.

The promotion rule required both sharp-change recall and precision to be at least as high as GRU's, in addition to positive paired RMSE gains. XGBoost missed the precision condition (52.82% versus 57.46%), so GRU remained the default next-day model despite XGBoost's lower average error. This decision follows the predeclared rule rather than choosing a model after viewing only its best metric.

B.3 Why attention accuracy is not explanation faithfulness

Spatial attention improved the spatial-patch model on its own 2023 test, but FA and TA did not improve the sequence model. Learned weighting can therefore help one architecture without reliably helping another. The faithfulness audit further separates accuracy from interpretability: attention rankings agreed less with replacement sensitivity than SHAP in all four comparisons. For S-SA under mean replacement, the highest-attention group's deletion changed output by 72.53 AQI points, close to 70.46 for a random ranking; SHAP's top group changed it by 73.76. This weak separation limits an explanation claim for that deletion test.

Replacement is an intervention on a fixed model, not a controlled experiment on atmospheric causes. Correlated inputs and unrealistic mean replacements can change rankings; the different same-city seasonal-donor correlations in Table 5 demonstrate that sensitivity. Thus, neither an attention heat map nor one perturbation score establishes the physical source of pollution.

C. Inference

The results address the project's central gap by evaluating prediction skill and explanation faithfulness separately. Recent AQI is a strong baseline; history-aware and weather-assisted models improve historical average error within their own paired cohorts. The improvements occur across all eight represented cities in the 2025 GRU and 2026 XGBoost comparisons, but unequal station availability and city sample sizes prevent a claim of uniform city-level reliability or transfer to an unseen city.

For the attention question, the supported conclusion is narrower: spatial attention improved one research architecture, while attention weights were less aligned with fixed-model sensitivity than SHAP on the sampled audits. Good prediction accuracy does not, by itself, validate attention as a faithful or causal explanation. A separate held-out-city evaluation would be needed to establish cross-city transfer of the attention research models.

The 30% GRU / 70% XGBoost blend was selected using earlier development data, but it has no independent prospective score and is excluded from the validated tables. Future evaluation requires forecasts captured before their target day and complete, quality-checked subsequent observations. Until then, the frozen GRU is the justified default for next-day forecasting.